

Homework 1

Due: Sept. 24 2026

1. Do Exercise 1 on page 7.
2. Consider a boundary-value problem (i.e. strong-form problem) with the Dirichlet boundary conditions imposed on both ends, that is,

$$(S) \left\{ \begin{array}{l} u_{,xx} + f = 0, \quad \text{in } (0,1) \\ u(0) = g, \\ u_{,x}(1) = h. \end{array} \right.$$

- (a) State its corresponding weak-form problem.
- (b) Show the equivalence between the strong- and weak-form problems.